Relational Data Model and Languages

Relational Data Model Concepts

- Relational Model in DBMS Relational model can represent as a table with columns and rows. Each row is known as a tuple. Each table of the column has a name or attribute.
- **Domain:** It contains a set of atomic values that an attribute can take.
- **Attribute:** It contains the name of a column in a particular table. Each attribute Ai must have a domain, dom(Ai)
- **Relational instance:** In the relational database system, the relational instance is represented by a finite set of tuples. Relation instances do not have duplicate tuples.
- **Relational schema:** A relational schema contains the name of the relation and name of all columns or attributes.
- **Relational key:** In the relational key, each row has one or more attributes. It can identify the row in the relation uniquely.

Relational Data Model Concepts

Example: STUDENT Relation

NAME	ROLL_NO	PHONE_NO	ADDRESS	AGE
Ram	14795	7305758992	Noida	24
Shyam	12839	9026288936	Delhi	35
Laxman	33289	8583287182	Gurugram	20
Mahesh	27857	7086819134	Ghaziabad	27
Ganesh	17282	9028 9i3988	Delhi	40

In the given table, NAME, ROLL_NO, PHONE_NO, ADDRESS, and AGE are the attributes.

The instance of schema STUDENT has 5 tuples.

t3 = <Laxman, 33289, 8583287182, Gurugram, 20>

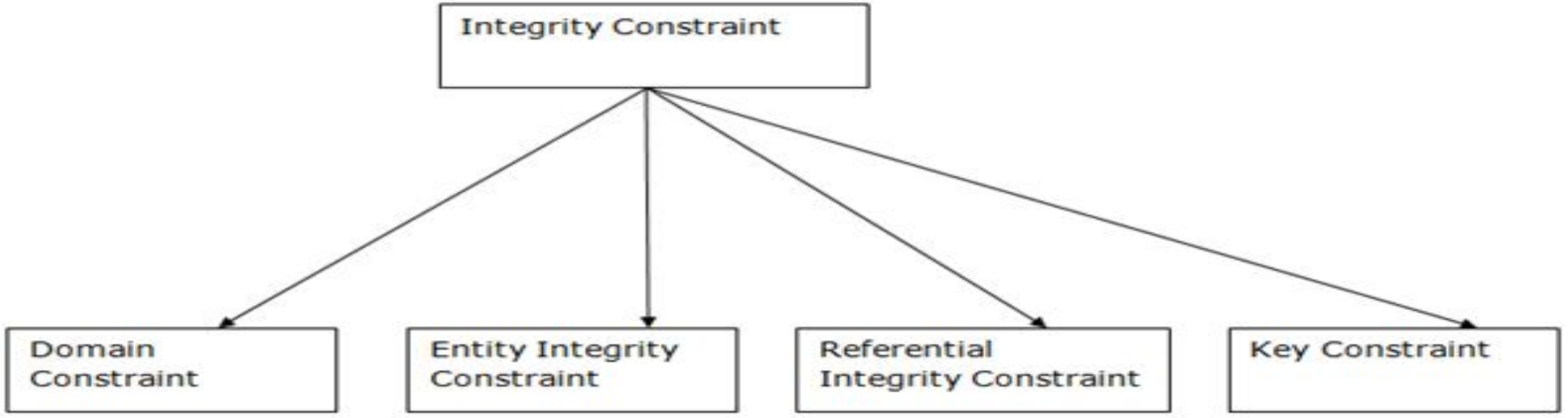
Relational Data Model Concepts

Properties of Relations

- Name of the relation is distinct from all other relations.
- Each relation cell contains exactly one atomic (single) value
- Each attribute contains a distinct name
- Attribute domain has no significance
- tuple has no duplicate value
- Order of tuple can have a different sequence

Integrity Constraints

- Integrity constraints are a set of rules. It is used to maintain the quality of information.
- Integrity constraints ensure that the data insertion, updating, and other processes have to be performed in such a way that data integrity is not affected.
- Thus, integrity constraint is used to guard against accidental damage to the database.



Domain constraints

- Domain constraints can be defined as the definition of a valid set of values for an attribute.
- The data type of domain includes string, character, integer, time, date, currency, etc. The value of the attribute must be available in the corresponding domain.

ID	NAME	SEMENSTER	AGE
1000	Tom	1st	17
1001	Johnson	2nd	24
1002	Leonardo	5th	21
1003	Kate	3rd	19
1004	Morgan	8th	A

Not allowed. Because AGE is an integer attribute

Entity integrity constraints

- The entity integrity constraint states that primary key value can't be null.
- This is because the primary key value is used to identify individual rows in relation and if the primary key has a null value, then we can't identify those rows.
- A table can contain a null value other than the primary key field.

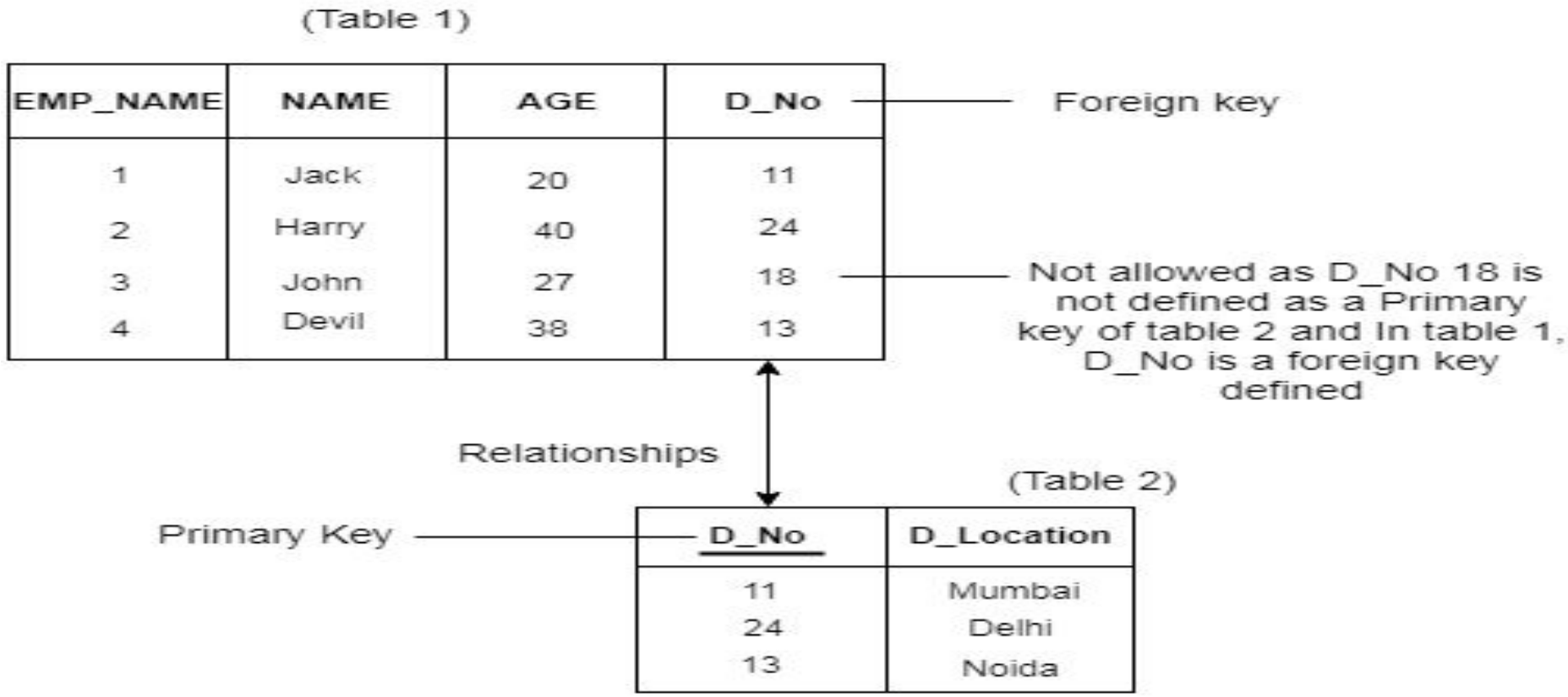
EMPLOYEE

EMP_ID	EMP_NAME	SALARY
123	Jack	30000
142	Harry	60000
164	John	20000
	Jackson	27000

Not allowed as primary key can't contain a NULL value

Referential Integrity Constraints

- A referential integrity constraint is specified between two tables.
- In the Referential integrity constraints, if a foreign key in Table 1 refers to the Primary Key of Table 2, then every value of the Foreign Key in Table 1 must be null or be available in Table 2.



Key constraints

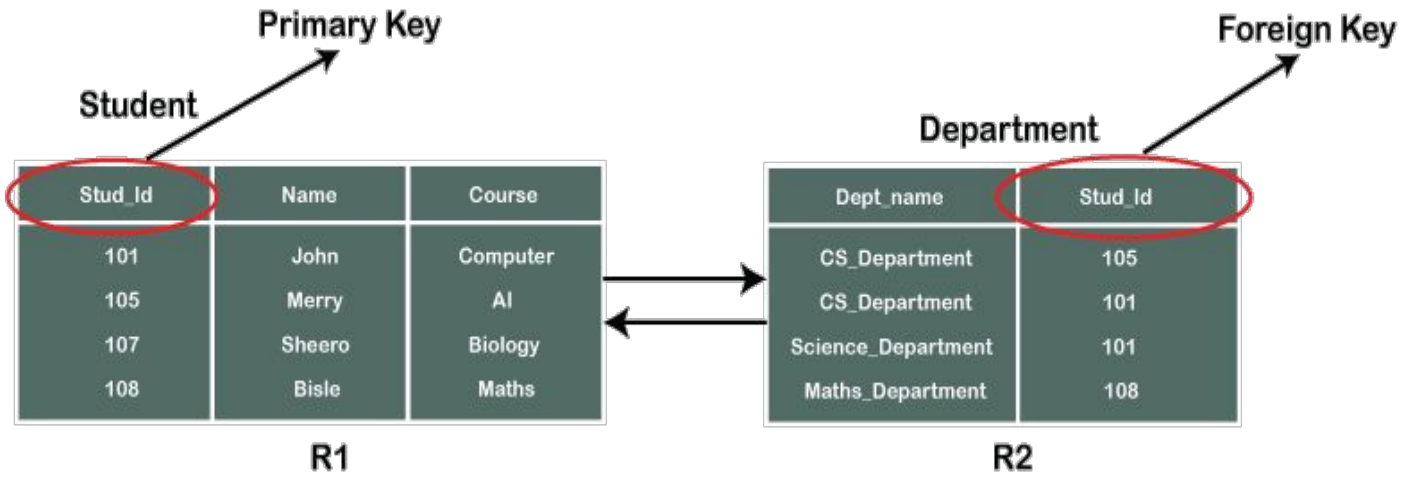
- Keys are the entity set that is used to identify an entity within its entity set uniquely.
- An entity set can have multiple keys, but out of which one key will be the primary key. A primary key can contain a unique and null value in the relational table.

ID	NAME	SEMENSTER	AGE
1000	Tom	1st	17
1001	Johnson	2nd	24
1002	Leonardo	5th	21
1003	Kate	3rd	19
1002	Morgan	8th	22

Not allowed. Because all row must be unique

Foreign Key

- A foreign key is different from a super key, candidate key or primary key because a foreign key is the one that is used to link two tables together or create connectivity between the two.
- A foreign key is the one that is used to link two tables together via the primary key. It means the columns of one table points to the primary key attribute of the other table. It further means that if any attribute is set as a primary key attribute will work in another table as a foreign key attribute. But one should know that a foreign key has nothing to do with the primary key.



Note: Referential Integrity in DBMS is developed from the concept of the foreign key. It is clear that a primary key is an alone existing key and a foreign key always reference to a primary key in some other table, in which the table that contains the primary key is known as the referenced table or parent table for the other table that is having the foreign key.

Candidate Key

- A candidate key is a set of attributes (or attributes) that uniquely identify the tuples in relation to or table. As we know the Primary key is a minimal super key, so there is one and only one primary key in any relationship but there is more than one candidate key that can take place. The candidate key's attributes can contain a NULL value which opposes to the primary key.

Super Key

- Super Key: Super Key is an attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation. All super keys can’t be candidate keys but the reverse is true.

Primary Key

- A Primary Key is the minimal set of attributes of a table that has the task to uniquely identify the rows, or we can say the tuples of the given particular table.

- The most significant advantages of a primary key are:
  - It uniquely identifies each row of a table
  - It gets a unique index for each primary key column that helps with faster access

Super Key	Candidate Key
Super Key is an attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.	Candidate Key is a subset of a super key.
All super keys can't be candidate keys.	But all candidate keys are super keys.
Various super keys together makes the criteria to select the candidate keys.	Various candidate keys together makes the criteria to select the primary keys.
In a relation, number of super keys is more than number of candidate keys.	While in a relation, number of candidate keys are less than number of super keys.
Super key attributes can contain NULL values.	Candidate key attributes can also contain NULL values.

- 1. Which of the following best describes a relation in the relational data model?
  A. A collection of programs
  B. A table consisting of rows and columns
  C. A collection of files
  D. A network of objects
- 2. In a relational database, a row is also known as a:
  A. Attribute
  B. Domain
  C. Tuple
  D. Relation
- 3. Which of the following represents a column in a relation?
  A. Tuple
  B. Attribute
  C. Record
  D. Key
- 4. Which constraint ensures that a primary key cannot contain NULL values?
  A. Domain Constraint
  B. Referential Integrity
  C. Entity Integrity
  D. Foreign Key Constraint
- 5. The entity integrity constraint states that:
  A. Foreign keys must be unique
  B. Primary key values cannot be NULL
  C. All attributes must contain numbers
  D. Every table must have two keys
- 6. Which constraint ensures that a foreign key value must either match a referenced key value or be NULL (where NULL is permitted)?
  A. Entity Integrity
  B. Domain Constraint
  C. Referential Integrity
  D. Key Constraint

7. A primary key is used to:
A. Store duplicate records
B. Uniquely identify each tuple in a relation
C. Connect only two databases
D. Define the data type of attributes

8. Which of the following can contain NULL values?
A. Primary Key
B. Candidate Key when selected as primary key
C. Foreign Key, if permitted by the schema
D. All keys must always contain values

9. A candidate key is:
A. Any attribute in a table
B. A minimal super key
C. Always a foreign key
D. A key that contains duplicate values

10. Which of the following is a super key?
A. A set of attributes that uniquely identifies tuples
B. Only the primary key
C. Only a foreign key
D. An attribute that allows NULL values

11. What is the main difference between a candidate key and a super key?
A. A candidate key may contain duplicate values
B. A candidate key is a minimal super key
C. A super key cannot uniquely identify a tuple
D. There is no difference

12. If a relation has multiple candidate keys, one of them is selected as the:
A. Foreign Key
B. Alternate Attribute
C. Primary Key
D. Domain Key

13. A candidate key that is not selected as the primary key is commonly called a(n):
A. Foreign Key
B. Alternate Key
C. Super Key
D. Domain Key

14. A foreign key is an attribute or set of attributes that:
A. Must always be the primary key of the same table
B. References a key in another or the same relation
C. Can never contain NULL
D. Must contain unique values

15. Consider STUDENT(Student_ID, Name, Dept_ID), where Student_ID is the primary key. If Dept_ID references DEPARTMENT(Dept_ID), then Dept_ID in STUDENT is a:
A. Candidate Key
B. Super Key
C. Foreign Key
D. Primary Key

16. Which constraint restricts the values that an attribute can take based on its defined domain?
A. Entity Integrity
B. Referential Integrity
C. Domain Constraint
D. Key Constraint

17. Which statement best describes a relation in the relational data model?
A) A collection of unordered tuples having the same attributes
B) A collection of attributes with different data types
C) A collection of keys without any constraints
D) A collection of tables connected by foreign keys

18. Consider the relation EMPLOYEE(Emp_ID, Name, Dept_ID, Salary). If Emp_ID uniquely identifies every employee, which constraint is primarily satisfied?
A) Referential integrity
B) Entity integrity
C) Domain constraint
D) Foreign key constraint

19. Which of the following is a valid characteristic of a primary key?
A) It may contain duplicate values
B) It may contain NULL values
C) It uniquely identifies each tuple
D) It must always consist of multiple attributes